



**WELCOME BACK
&
THANK YOU**



**<Creative
Software/>**

**Advance Java
Crash Course**

For TD Bank



MEET YOUR CRASH COURSE TEAM

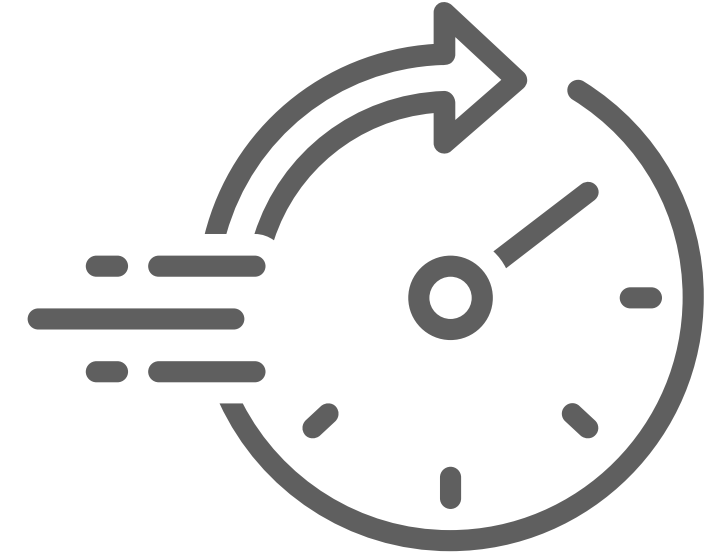


TANGY F.
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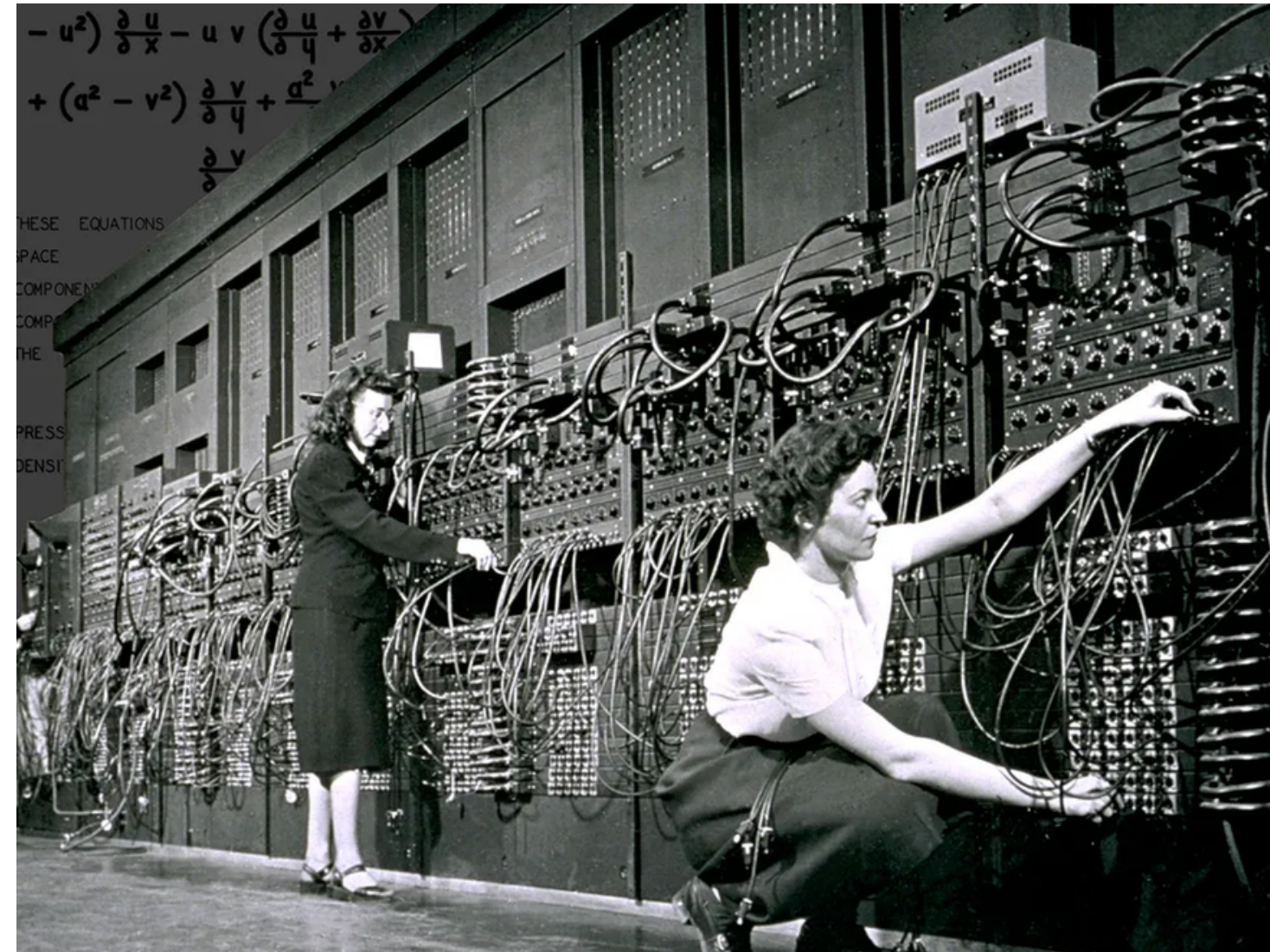
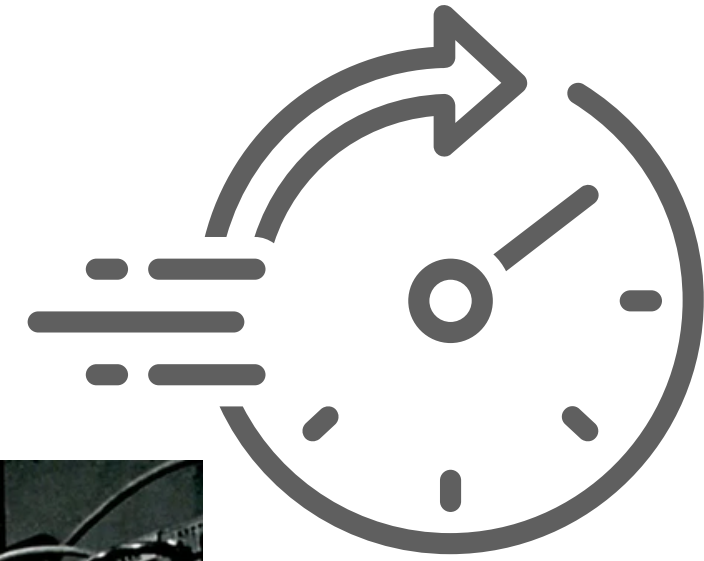
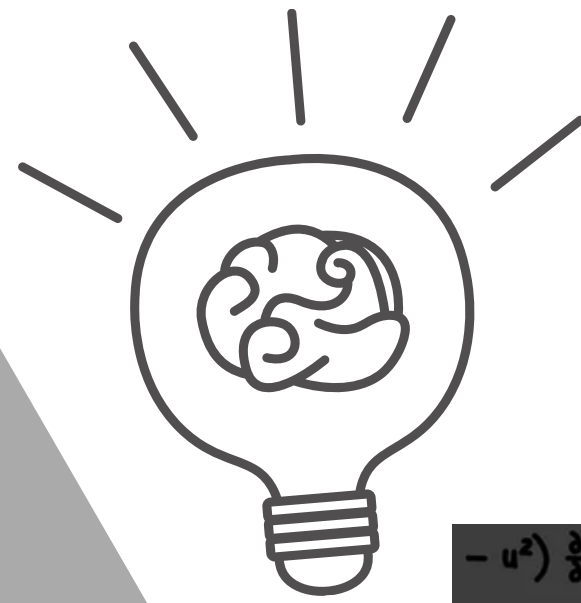
Rapid Review **TRIVIA**



Rapid Trivia

What was the name of the first digital computer?

Rapid Review **TRIVIA**



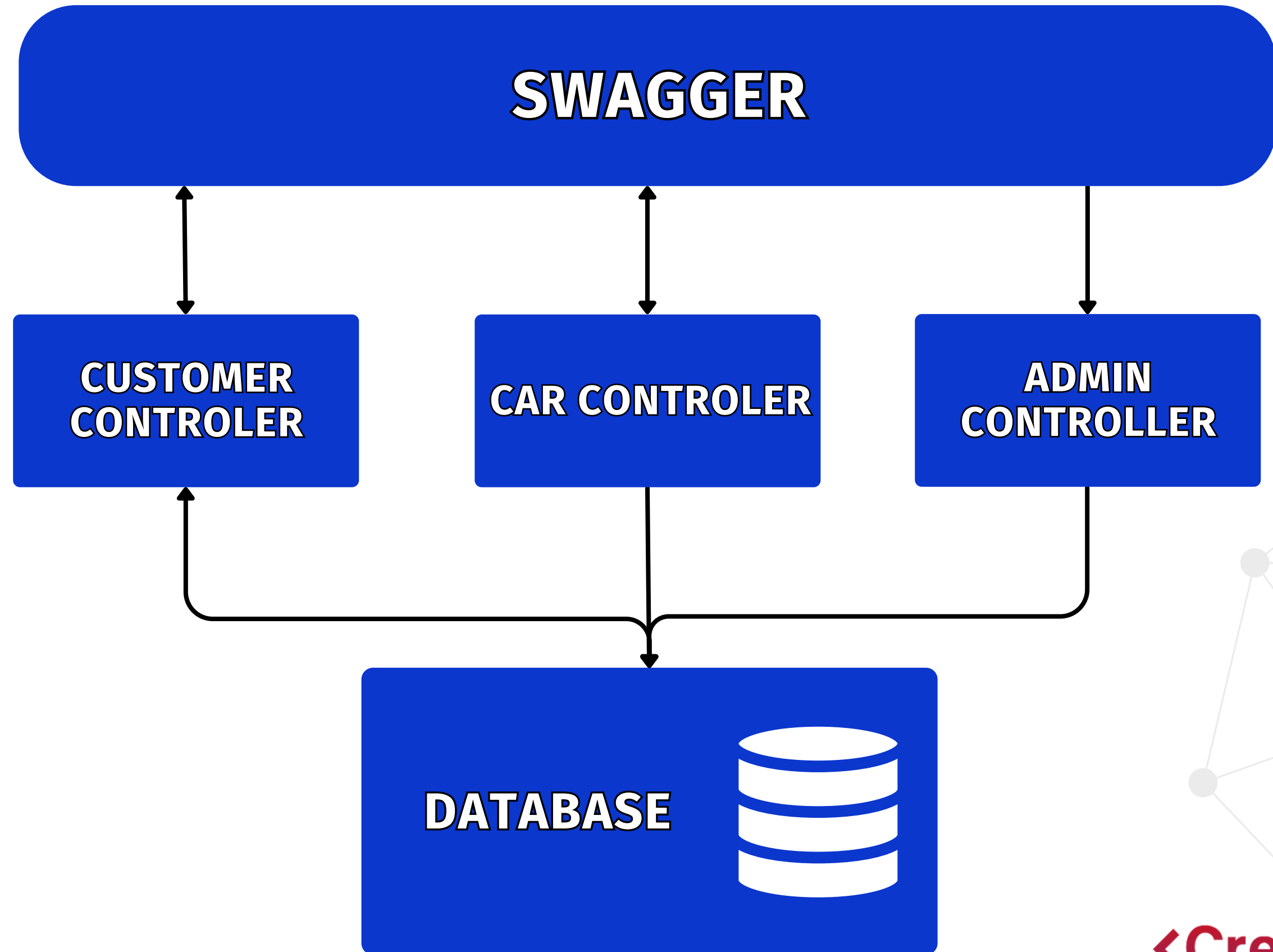
ENIAC Electronic Numerical Integrator and Computer

TRAINING DAY 10

TODAY'S AGENDA

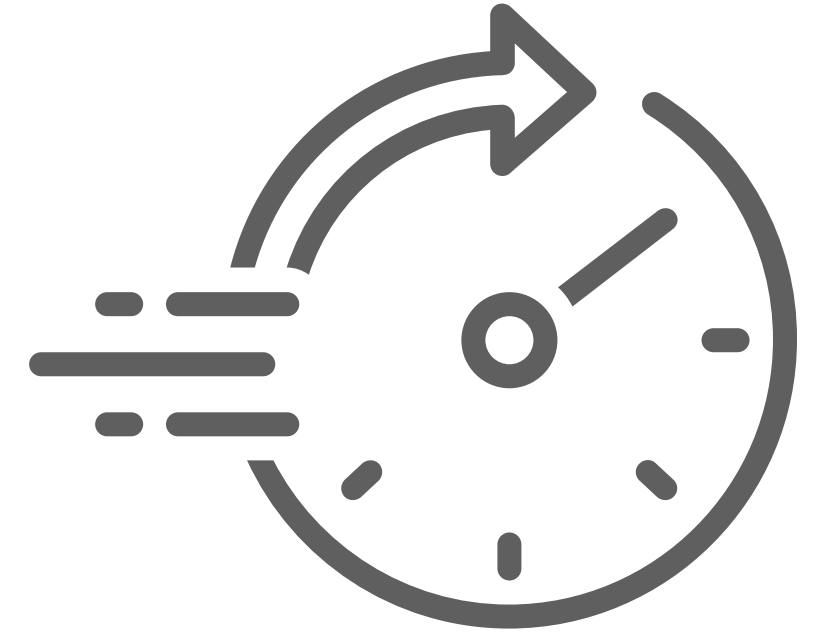
- 1 Yesterday's Coding Exercise to Go Rapid Review
- 2 12 Factor App
- 3 Case Study. Using Java Reflection in a Corporate EDI Framework
- 4 Post Assessment Questions

WE ARE BUILDING



**BITE SIZE
CAR RENTAL APP**

Rapid Review Lesson 1



Rapid Review

Add Introspection to car-controller `/api/cars/{carId}`

CODING EXERCISE TO GO



In the **car-controller** `/api/cars/{carId}` change the line bellow to call method **getCarById(carId)** via introspection.
Use a String variable to store "getCarById"

```
CarResponseDto car = carService.getCarById(carId);
```

GET

`/api/cars/{carId}`

YESTERDAY'S CODING TO GO REVIEW



```
@GetMapping("/{carId}")
public ResponseEntity<CarResponseDto> getCarById(@PathVariable Long carId) throws Exception {
    Class<?> clazz = CarService.class;
    String methodName = "getCarById";
    Method method = clazz.getDeclaredMethod(methodName, Long.class);
    CarResponseDto car = (CarResponseDto) method.invoke(carService, carId);

    //CarResponseDto car = carService.getCarById(carId);
}
```


DAY 10
LESSON 2

12 Factor App

1. Codebase

>>Version control
best practice

Example: Git

2. Dependencies

>>Using a dependency
management system
like Maven or Gradle
>>Spring Boot provides
dependency
management plugins

3. Config

>>Have all configuration in
properties or any YAML file
instead of hard coding them.
>>Using environmental
variable and external config
files

4. Backing Services

- >>DB or external services are considered as backing services
- >>Spring boot provides integration modules and libraries such as JPA, Hibernate for connection

5. Build, Release, Run

- >>Use Maven or Gradle for build application into executable JAR or WAR files
- >>Spring supports these executable file to easily run and deploy.

6. Processes

- >> Designing a stateless process and having the data store in DB or in Caches
- >> Spring supports the DB storage and external cache service

7. Port Binding

- >> Bind the services or Rest applications to specific port
- >> In spring we can adjust the port number for Tomcat or Jetty

8. Concurrency

- >> Scaling the applications horizontally
- >> Spring gives the content caching or distributed caching to handle concurrency

9. Disposability

- >>Ensure fast startup and graceful shut down of application
- >>Spring provides lifecycle hooks and shutdown events to cleanup activity.

11. Logs

- >>Having logging enabled for the application. It helps in monitoring and trouble shooting.
- >>Spring Boot has a built-in logging framework like Log4j

10. Dev/ Prod Parity

- >>Maintaining all the environments as possible.
- >>Spring Profiles can be used to manage environment specific configs.

12. Admin Processes

- >>One-off processes should be performed using the same environment and tools as the production environment.
- >>Spring boot features like Spring Batch and scheduling tasks to execute the administrative processes.



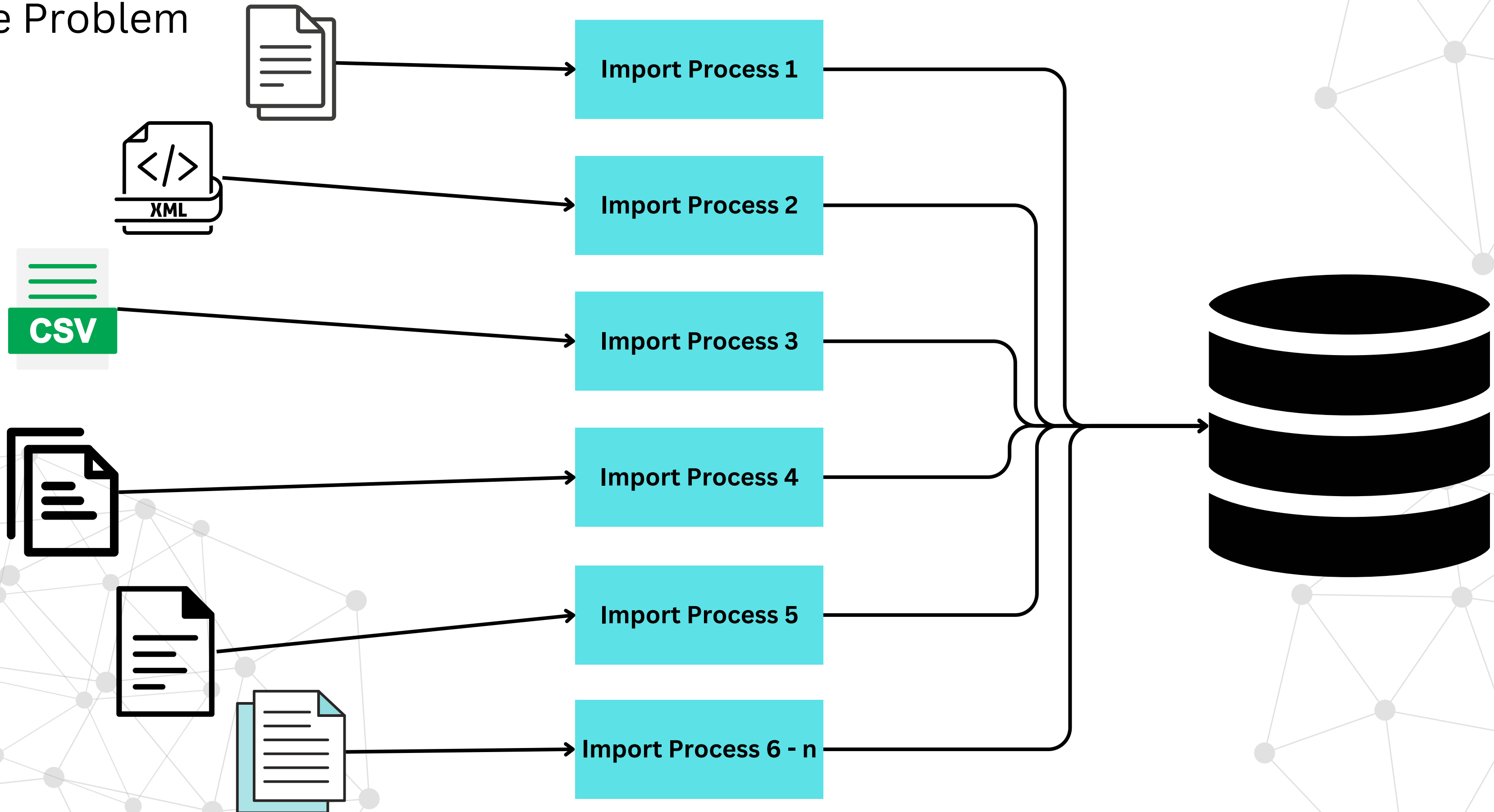
DAY 10

LESSON 3

Case Study. Using Java Reflection in a Corporate EDI Framework

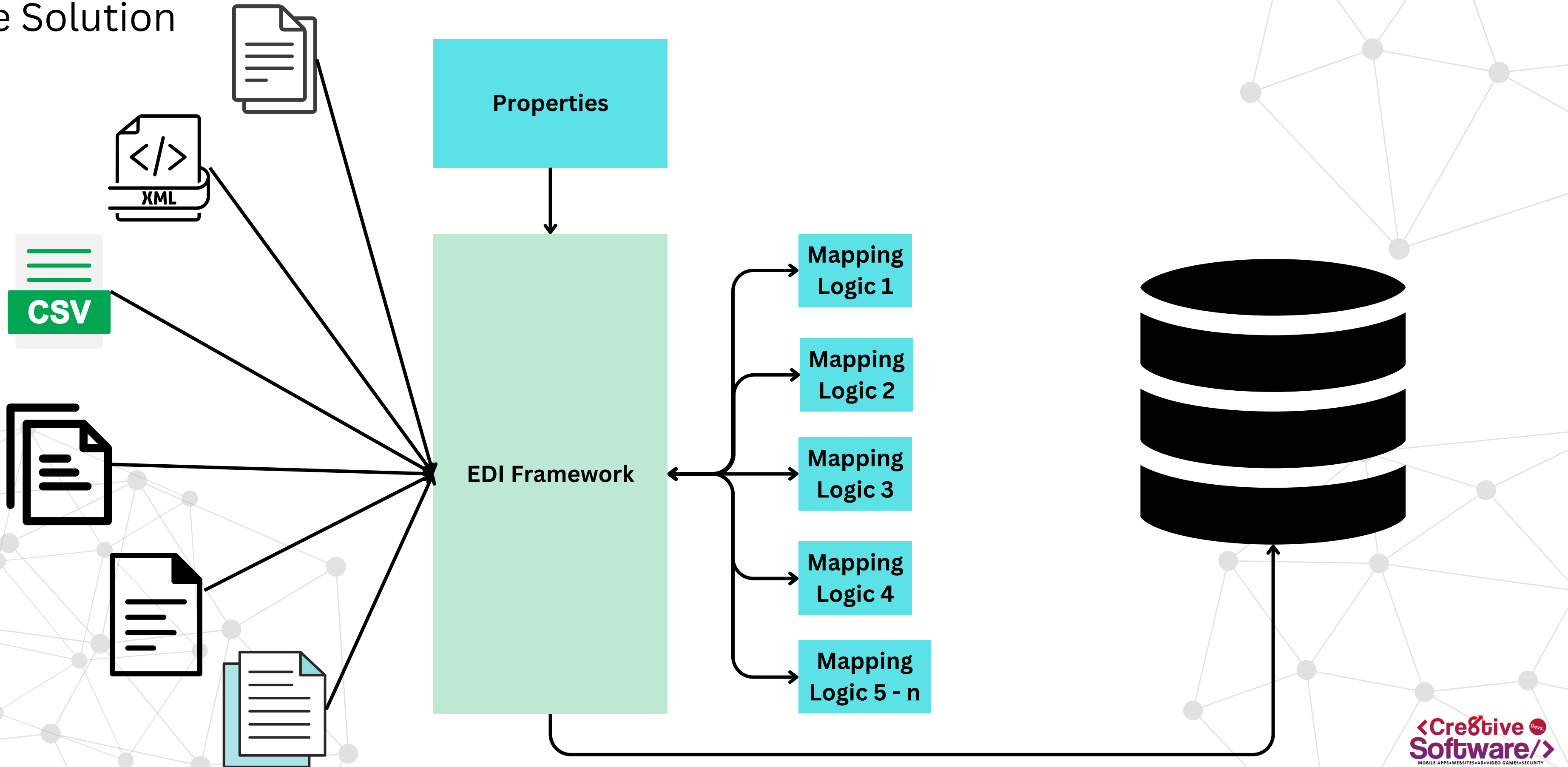
CASE STUDY USING JAVA REFLECTION

The Problem

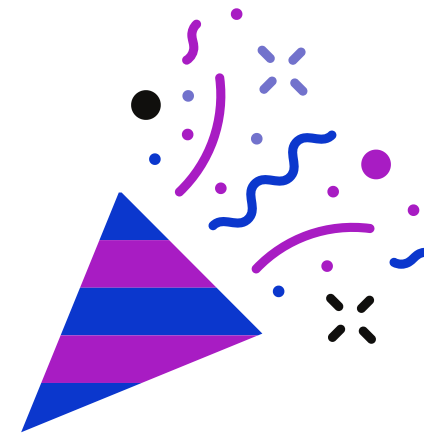


CASE STUDY USING JAVA REFLECTION

The Solution



LESSON 10



**Type your answers on
a document
& email them to us.**

QUESTIONS 1 - 3

LESSON 10



1. We are getting a compile error, it seems like a dependency is missing. What is the Gradle command that we should use to know for sure?
2. What should we do with the the **bookVehicleDto** object when we are sure we no longer need it and we want the garbage collector to remove it from memory.
3. Which of the 2 approaches bellow will yield the best performance, the first or the second.

// First Logic

```
String value = "";  
for (int i = 0; i < n; i++) {  
    value += i;  
}
```

// Second Logic

```
StringBuilder stringBuilder = new StringBuilder();  
for (int i = 0; i < n; i++) {  
    stringBuilder.append(i);  
}
```

QUESTIONS 4

LESSON 10



4. We know **Cat** extends **Domestic**, and **catList** is a **List** of **Cat**. If we have the call **setDomestic(catList)**; which of the implementations bellow will NOT give a compilation error to such call.

// Implementation 1

```
private void setDomestic(List <? extends Domestic> domesticList);
```

// Implementation 2

```
private void setDomestic(List <Domestic> domesticList);
```

QUESTIONS 5 - 6

LESSON 10



5.The Lambda expression bellow is not compiling, what is wrong with it?

```
FunctionalCalculator functionalCalculator -> (num1, num2) = num1 * num2;
```

6.We are getting a **ClassNotFoundException** when we run the code bellow despite “**TheClass**” existing in the package. What is missing?

```
String className = "TheClass";  
Class<?> clazz = Class.forName(className);  
ClassInterface instance = (ClassInterface) clazz.getDeclaredConstructor().newInstance();
```



<Creative Devs.
Software/>



**HEARTFELT GRATITUDE
FOR YOUR PARTICIPATION
FROM
CRE8TIVE DEVS**

**CERTIFICATE
OF
COMPLETION**





THANK YOU

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Crash Course

We will see you tomorrow

